

ANKITH REDDY SUBHANPURAM

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SUMMARY

ML Engineer and AI Researcher with hands-on experience building production-grade LLM systems, RAG pipelines, and fine-tuned foundation models across clinical, cloud, and defense domains. Proficient in end-to-end MLOps on AWS, Azure, and HPC clusters; currently pursuing an M.S. in AI Systems at the University of Florida (GPA: 3.89).

PROFESSIONAL EXPERIENCE

University of Florida – Intelligent Clinical Care Center (IC3 / PRISMAp)

Gainesville, FL

Research Assistant

Research | Aug 2025 – Present

Research I: Clinical Digital Twin – Synthetic Q&A Generation & LLM Fine-tuning

- Designed and iterated multi-stage prompting strategies using frontier models to synthesize high-fidelity clinical question-answer pairs from EHR data, generating large-scale training corpora for downstream fine-tuning.
- Evaluated prompt quality systematically using DeepEval with 14 LLM-as-judge metrics covering clinical reasoning, factual accuracy, and diagnostic validity across 110K+ EHR samples.
- Applied semantic deduplication on the generated Q&A corpus using ClinicalBERT and PubMedBERT embeddings with FAISS + Jaccard pipelines, reducing dataset redundancy by 36% and improving downstream model accuracy by 4%.
- Fine-tuned GPT-based OSS 120B and 20B models and LLaMA-3.3 70B using Axolotl with PEFT/LoRA on HiPerGator's NVIDIA B200 GPU cluster leveraging FSDP2 for fully sharded, parallelized training.
- Benchmarked fine-tuned models against clinical OSS baselines (AlpaCare, ChatDoctor) using 3 frontier LLM judges for response quality assessment, and evaluated on public clinical NLP benchmarks – MedQA, HeadQA, and PubMedQA – to validate diagnostic reasoning and knowledge generalization.

Research II: ICU EHR Deep Learning – Transformer, Mamba & Mixture-of-Experts Modeling

- Engineered multimodal ICU EHR datasets by preprocessing preoperative, intraoperative, and postoperative static and temporal time-series data alongside unstructured clinical notes into structured HDF5 (.h5) files optimized for deep learning ingestion.
- Trained and benchmarked Transformer, Mamba (state space model), and Mixture-of-Experts (MoE) architectures on longitudinal ICU patient data to model complex clinical trajectories; currently in the grant evaluation phase.

Research III: Pathology Report Analysis

- Developing advanced prompting techniques (chain-of-thought, few-shot, structured extraction) with frontier LLMs for automated analysis and structured information extraction from unstructured pathology reports.

University of Florida – IPPD | Oelrich Construction

Gainesville, FL

AI Engineer

Co-op | Aug 2025 – Present

- Architected a permissions-aware enterprise RAG system enforcing Role-Based Access Control (RBAC) at the document level by integrating Azure Active Directory (Azure AD) with the Microsoft Graph API to programmatically extract per-file access permissions from Azure file shares.
- Built a document ingestion pipeline using SuryaOCR and PyMuPDF for multi-format extraction; generated vector embeddings via Azure OpenAI embedding models and implemented SHA-based content hashing on chunks and RBAC metadata for idempotent ingestion, eliminating redundant nightly reprocessing.
- Engineered a hybrid retrieval system fusing BM25 sparse keyword search with DiskANN-indexed dense vector similarity search, followed by cross-encoder reranking, delivering precision-optimized, permission-scoped results to authenticated users.
- Designed a LangGraph and LangChain chunking and indexing workflow persisting chunk embeddings alongside RBAC access policies into Azure Cosmos DB (NoSQL); secured all runtime credentials via Azure Key Vault.
- Containerized the ingestion pipeline as an Azure Container Apps Job with scheduled nightly execution for zero-touch data refresh; deployed a FastAPI backend on Azure App Services with server-side session management and Azure OpenAI-powered conversational retrieval.

Scholarship Auditions

Nashville, TN

Artificial Intelligence Engineer

Internship | May 2025 – Aug 2025

- Architected a production-grade RAG chatbot on AWS ECS/Fargate using FastAPI and LangGraph orchestration, backed by Amazon OpenSearch Service as the vector store and Amazon Bedrock for context-aware LLM inference across 20K+ documents.
- Implemented hybrid retrieval combining dense embedding search and cross-encoder reranking, improving retrieval relevance by 25% and reducing end-to-end response latency by 35%; deployed session memory via Amazon ElastiCache (Redis) with distributed rate limiting across ECS replicas.
- Built access-control middleware with JWT authentication and tenant isolation; secured runtime credentials via AWS Secrets Manager and enforced ALB + WAF ingress controls with structured CloudWatch Logs observability.
- Fine-tuned ResNet-50 for music transcription image classification via transfer learning, achieving 90%+ accuracy on 10K+ annotated images; containerized services with Docker and Amazon ECR for zero-downtime rolling deployments on Fargate.
- Constructed automated web-scraping pipelines using Scrapy and Selenium to collect and structure data from 50+ sources, generating 100K+ records; orchestrated workflows via Apache Airflow for scheduling and fault-tolerant execution.

Centre for Artificial Intelligence and Robotics (DRDO)

Bengaluru, India

Artificial Intelligence Research Intern

Research Internship | Oct 2023 – Mar 2024

- Designed and pre-trained a Conformer model on restricted defense speech corpora using self-supervised contrastive learning objectives, then coupled it with Wav2Vec2 as a feature extractor; executed multi-GPU data-parallel training (DDP) in PyTorch on NVIDIA DGX hardware, reducing overall training time by 40% on 1K+ hours of speech data.
- Fine-tuned the Conformer-Wav2Vec2 architecture end-to-end using Connectionist Temporal Classification (CTC) as the training objective, with Byte Pair Encoding (BPE) tokenization for subword vocabulary construction; improved recognition for low-resource languages (Assamese, Bengali), achieving 8.3% WER and reducing vocabulary size by 30%.
- Deployed Voice Activity Detection (VAD) and noise-reduction pipelines in production, increasing transcription accuracy by 15% in real-world noisy environments.

SKILLS

Programming: Python, SQL
AI/ML: Machine Learning, Deep Learning, NLP, LLMs, RAG Systems, LLM Fine-tuning (LoRA, QLoRA, PEFT), Prompt Engineering (CoT, Few-Shot), Hybrid Retrieval (BM25 + Dense), Reranking, GenAI, Computer Vision, Speech Recognition, Conformer, Contrastive Learning, CTC, Mamba (SSM), Mixture-of-Experts (MoE)
Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, LangGraph, FastAPI, Gradio, Scikit-Learn, Pandas, NumPy, Axolotl, DeepEval, FAISS, DiskANN, PyMuPDF, SuryaOCR, Librosa, NLTK, SpaCy
AWS: ECS/Fargate, ECR, ElastiCache (Redis), OpenSearch Service, Bedrock, Secrets Manager, CloudWatch, ALB, WAF, VPC, IAM
Azure: Azure OpenAI, Azure Active Directory (AD), Microsoft Graph API, Cosmos DB, Key Vault, Container Apps Jobs, App Services, Azure File Share
MLOps & HPC: MLflow, FSDP2, Distributed Data Parallel (DDP), NVIDIA DGX/B200, HiPerGator, Apache Airflow, Docker, Terraform, Git
Data & Web: HDF5, Data Analytics, Web Scraping, Scrapy, Selenium

PROJECTS

OrchesterAI

- Architected a stateful multi-agent orchestration framework (LangGraph, LangChain) coordinating 7 specialized LLM agents via a DAG with autonomous conditional edge routing and type-safe inter-agent schema validation.
- Built an autonomous email & calendar execution layer (FastAPI, Gradio, LangSmith, Gmail/Outlook/Calendar APIs) with OAuth 2.0 authentication, fuzzy-match contact resolution, and end-to-end distributed tracing for full agent observability.

Virtual Surya Namaskar Coach

- Developed a real-time yoga posture analysis system (OpenCV, TensorFlow, MediaPipe) using full-body landmark tracking and joint-angle extraction to classify 12 Surya Namaskar poses, achieving sub-100ms inference latency on live webcam input.
- Engineered a pose-correction pipeline with reference angle schemas and data augmentation, enabling per-joint deviation detection that flags incorrect form in real time.

EDUCATION

University of Florida	Gainesville, FL
Master of Science in Artificial Intelligence Systems	Aug 2024 – May 2026
Cumulative GPA: 3.89 / 4.00	
Vignan’s Bharathi Institute of Technology	Hyderabad, Telangana, India
Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning)	Aug 2020 – May 2024
CGPA: 8.76 / 10.00	